

Subject Description Form

Subject Code	EE2029 / EE2029B
Subject Title	Transportation Engineering Fundamentals
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To introduce the fundamental topics and concepts of transportation engineering, including traffic engineering, transportation economics, public transportation, smart transportation, etc. 2. To explain the operations of real-life transportation systems and the related engineering, economics, and environmental issues. 3. To describe the basic techniques of transportation system analysis. 4. To equip students with a broad knowledge base in transportation engineering and sound practical sense, preparing them to choose the Transportation Systems Engineering degree path.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify the key issues in transportation systems. b. Appreciate the problems and suggest solutions to real-life transportation problems. c. Conduct basic system analysis. d. Understanding the prospects and new trends in the transportation industry. e. Be ready to study transportation-related subjects at a higher level.
Subject Synopsis/ Indicative Syllabus	1. Transportation systems: Introduction to transportation engineering, transportation problems and solutions, transportation systems in Hong Kong, transportation in social, economic, environmental, and political roles, and transportation modes. 2. Smart transportation: Intelligent transportation systems, connected and autonomous vehicles, electrified transportation, shared mobility, low-altitude economy, and data-driven methods and solutions. 3. Traffic engineering: Elements of traffic engineering, time-space diagram, speed-flow-density relationships, queueing theory, traffic measurement, level of service. 4. Public transportation (transit): Designs, management, and operations of public transportation systems; transit network structures and operation schemes; transit service reliability and adaptive control; demand-responsive service; transit technologies; rail transportation. 5. Transportation economics: Principles of transport economics; transportation demand and supply; transportation policy, e.g. pricing; transportation network analysis.
Teaching/ Learning Methodology	The key concepts and techniques covered in this subject are discussed in lectures and guest lectures, including guest lectures. Individual assignments consist of numerical problems, open-ended questions, mini projects, and a final term essay, which let students demonstrate their level of understanding and create evidence of learning.

	Learning/Learning Methodology		Outcomes				
			a	b	c	d	e
	Lectures		✓	✓	✓	✓	✓
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	1. Assignments (including mini projects)	60%	✓	✓	✓	✓	✓
	2. Midterm exam	25%	✓	✓	✓	✓	✓
	3. Final term essay	15%	✓	✓	✓	✓	✓
	Total	100%					
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The students will be assessed with three components: 4-5 written assignments, including problem sets (2) and mini projects (3), a midterm examination, and a final term essay. The written assignments will consist of numerical, descriptive, and real-system measurement, diagnosis, and design problems to address different aspects of skills required in achieving intended learning outcomes (a), (b), (c), (d), and (e). The midterm exam is conducted mid-semester (when most theoretical and quantitative contents have been taught) to consolidate students' knowledge in lectures, class activities, and self-learning activities. It appropriately assesses intended learning outcomes (a)-(d). The final term essay is conducted at the end of the semester to let students showcase and allow the lecturer to examine their knowledge learned about transportation systems and engineering practice, interests and contemplation on transportation issues, and motivations for choosing transportation engineering. It appropriately assesses intended learning outcomes (a)-(e).							
Student Study Effort Expected	Class contact:						
	▪ Lectures						39 Hrs.
	Other student study efforts:						
	▪ Reading and studying						46 Hrs.
	▪ Completion of assignments						20 Hrs.
	Total student study effort						105 Hrs.
Reading List and References	1. C.F. Daganzo, Fundamentals of Transportation and Traffic Operations, Pergamon, 2008. 2. C.F. Daganzo and Yanfeng Ouyang, Public Transportation Systems: Basic Principles of System Design, Operations Planning and Real-Time Control, 2019. 3. J. Sussman, Introduction to Transportation Systems, Boston: Artech House, 2000. 4. Jon D. Fricker and R.K. Whitford, Fundamentals of Transportation Engineering – A Multimodal Systems Approach. Prentice Hall, 2004. 5. E. Quinet and R. Vickerman, Principles of Transport Economics, Edward Elgar Publishing Limited, 2004.						